

DHCP — Guided Notes ANSWER KEY

Unit 1.6.1 | CompTIA Network+ N10-009 Obj. 1.6 | N10-008 1.6

For instructor use / distribute after students complete the notes.

Question 1

A DHCP scope defines the _____ of addresses available for assignment, the _____ time, and options such as _____, _____, and _____. Exclusions are addresses within the scope range that the server will _____.

Answer: Range; lease; default gateway, DNS server, domain name (any two); never assign.

Question 2

A DHCP reservation assigns a _____ IP address to a specific device based on its _____. This differs from a static IP because the address is still managed by _____ rather than configured manually on the device.

Answer: Permanent (fixed); MAC address; the DHCP server.

Question 3

When a DHCP lease is _____ % expired, the client attempts to renew with the same server. If the server is unavailable when the lease fully expires, the client falls back to _____ and starts the _____ process over.

Answer: 50%; APIPA (169.254.x.x); DORA.

Question 4

A DHCP relay agent (helper address) is needed when _____. The relay agent is configured on the _____ and forwards DHCP broadcasts as _____ packets to the DHCP server's IP address.

Answer: The DHCP server and client are on different subnets (routers don't forward broadcasts by default); the router/Layer 3 switch interface; unicast.

Question 5

DHCP failover has two modes. In _____ mode, one server handles all leases and the other takes over only if the primary fails. In _____ mode, both servers share the address pool and handle requests simultaneously.

Answer: Hot-standby; load-balance.

Question 6

Match each DHCP problem to its symptom:

Exhausted scope → _____

Rogue DHCP server → _____

Misconfigured relay → _____

IP address conflict → _____

Answer: Exhausted scope → new devices get APIPA addresses, existing leases still work; Rogue DHCP server → devices get wrong gateway/DNS, lose connectivity; Misconfigured relay → devices on remote subnets get APIPA; IP conflict → both devices show warnings, intermittent connectivity for one or both.

Question 7

To check available addresses in a Windows DHCP scope, you would use _____. To force a client to release and request a new lease, the commands are _____ followed by _____.

Answer: DHCP Manager console (or netsh/PowerShell); ipconfig /release; ipconfig /renew.

Question 8 — Real World Application

REAL WORLD: On Monday morning, 30 students in one classroom can't get on the network. They all have 169.254.x.x addresses. Students in other classrooms are fine. Walk through your diagnosis in order.

Answer: 1. APIPA on all devices in one segment = DHCP failure isolated to that subnet. 2. Check the DHCP scope for that subnet — is it exhausted? 3. Check the relay agent (helper address) on the switch or router interface for that VLAN — is it configured and pointing to the correct DHCP server? 4. Check if the DHCP server is running and the scope is active. 5. Check for a rogue DHCP server on that segment (packet capture or DHCP snooping logs). 6. Try ipconfig /release /renew on one workstation after each fix to confirm.